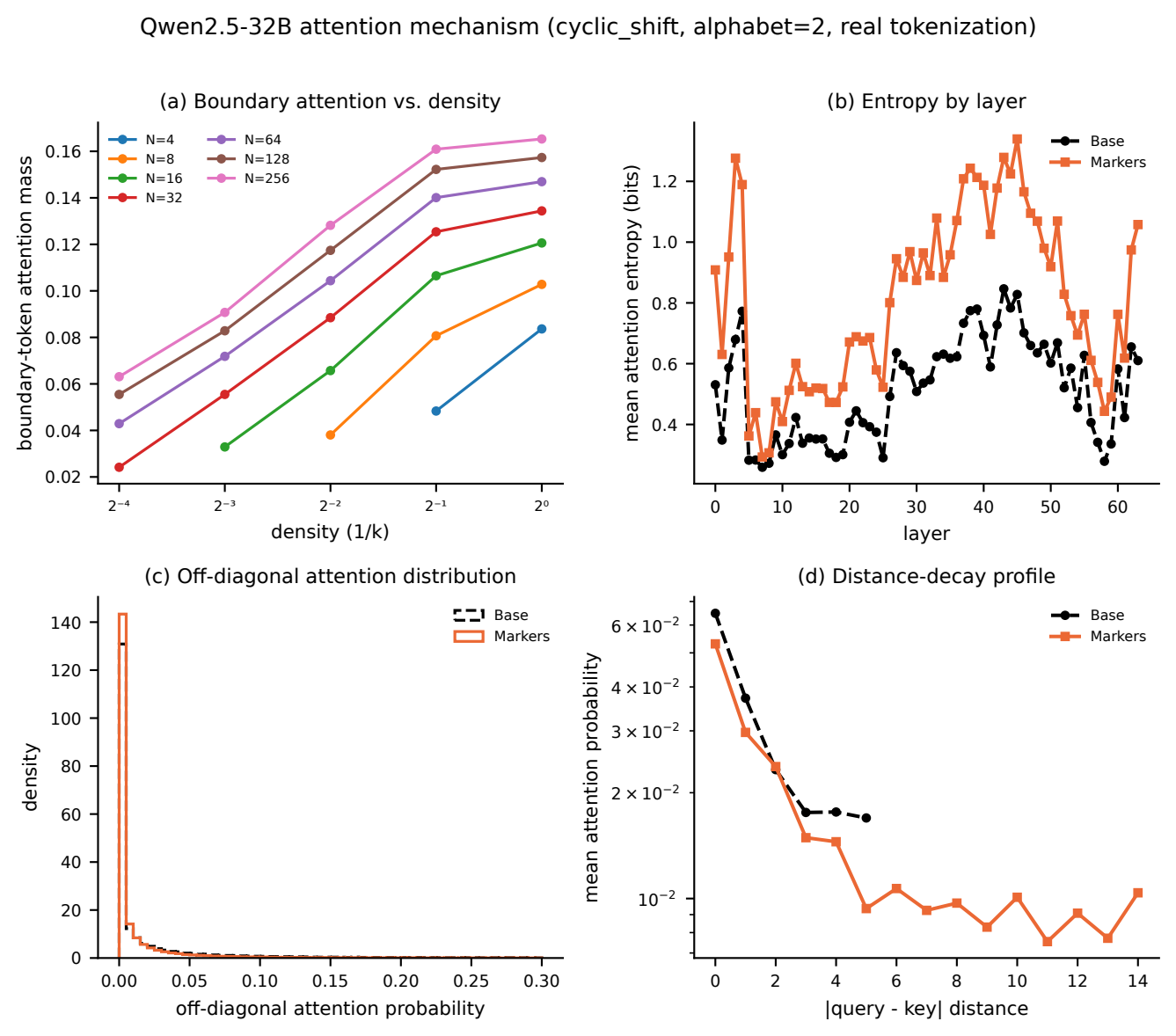

Qwen2.5-32B attention mechanism (cyclic_shift, alphabet=2, real tokenization)
(a) Boundary attention vs. density
boundary-token attention mass
density (1/k)
N=4
N=8
N=16
N=32
N=64
N=128
N=256
(b) Entropy by layer
mean attention entropy (bits)
layer
Base
Markers
(c) Off-diagonal attention distribution
density
off-diagonal attention probability
Base
Markers
(d) Distance-decay profile
mean attention probability
|query - key| distance
Base
Markers